



MSc Business Analytics

Module IS6611: Applied Research in Business Analytics

IT Artefact Version 3 Report

Project Title: Stock Market Manipulation Detection

A Business Analytics Framework for Market Surveillance and Compliance Decision Support

A Russell 2000 Case Study

Group 9

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Executive Summary

Every day, retail investors in small-cap US stocks trade under an informational disadvantage that the market is supposed to prevent. Pump-and-dump schemes, spoofing, and wash trading quietly distort prices in the Russell 2000 - the index of roughly 2,000 small-cap stocks where regulatory coverage is thinnest - and by the time a regulator is alerted, the damage to retail capital is already done. This project builds and delivers a surveillance system that flips that timeline: flagging high-risk trading activity while it is still forming, not after the crash.

The surveillance system fuses two complementary signals into a single 0–100 composite risk score. The primary signal is an Isolation Forest anomaly detector trained on 467,799 real 2024 Russell 2000 trading records, validated with $F1 = 0.7579$, $Precision = 0.75$, $Recall = 0.77$ and $Accuracy = 0.99$. The secondary signal, added on the industry mentor's guidance, is a FinBERT financial-news sentiment engine that detects manipulation attempts surfacing in company narratives before they appear in the price. Every flagged record is accompanied by a plain-language explanation - SHAP feature attribution for compliance analysts, LIME for traders - so that every alert can be justified, defended and escalated without ambiguity.

The third pillar of the system is operational: a prepared CSV dataset is placed in a designated input folder, which triggers an SSIS Execute Process Task that runs the Python analysis script. The script reads the CSV, computes all features, runs the Isolation Forest and FinBERT composite scoring, and writes an integrated output file that Power BI reads directly. Power BI auto-refreshes from that file - so from raw data to three live stakeholder dashboards, the chain runs unattended.

This report structures that account using the Data Value Map (D. Sammon, 2024) and closes by situating the results against their honest limits: where the evidence is strong, and where it stops short of full production deployment.

1. Introduction and Problem Statement

What Is Stock Market Manipulation and Why Does It Matter?

Stock markets work because prices are supposed to reflect genuine supply and demand. When manipulators distort those prices artificially, ordinary investors make decisions based on a fabricated reality and bear the resulting losses. The three most common tactics - illustrated in Figure 1 - are Pump and Dump (artificial inflation followed by a coordinated sell-off), Spoofing (placing and cancelling large orders to mislead other traders), and Wash Trading (buying and selling between controlled accounts to create false volume). Research confirms these are not edge cases: Putnins (2012) estimates a meaningful proportion of unusual small-cap trading activity has manipulation as its cause, with retail investors disproportionately bearing the losses.



Figure 1: The Threat Triad - Pump and Dump, Spoofing, and Wash Trading and their impact on retail investors (Group 9, 2026).

Why the Russell 2000?

The Russell 2000 tracks the 2,000 smallest listed US companies - thin-volume stocks with limited institutional oversight and concentrated regulatory attention elsewhere. Comerton-Forde and Putnins (2015) confirm significantly higher manipulation prevalence in small-caps; Imisiker and Tas (2016) identify thin liquidity as the primary enabling factor. The result is a structural surveillance gap: by the time a compliance officer investigates, the Setup–Pump–Dump cycle is already complete. Our problem statement: stock markets lack data-driven mechanisms to detect manipulation before retail capital is destroyed. Our response: a system that identifies convergent anomaly signals during the Setup phase and delivers an explainable, actionable alert to every stakeholder with zero manual steps between data and dashboard.

The AS-IS and TO-BE Transformation

Figure 2 captures the paradigm shift. In the AS-IS state (Figure 2a), compliance sits at the far right of the process - responding only after the Dump has occurred and losses are locked in. In the TO-BE state (Figure 2b), automated daily ingestion means every stock is scored every trading day. When the threshold is breached, an alert fires immediately and compliance acts during the Setup phase, not after it.

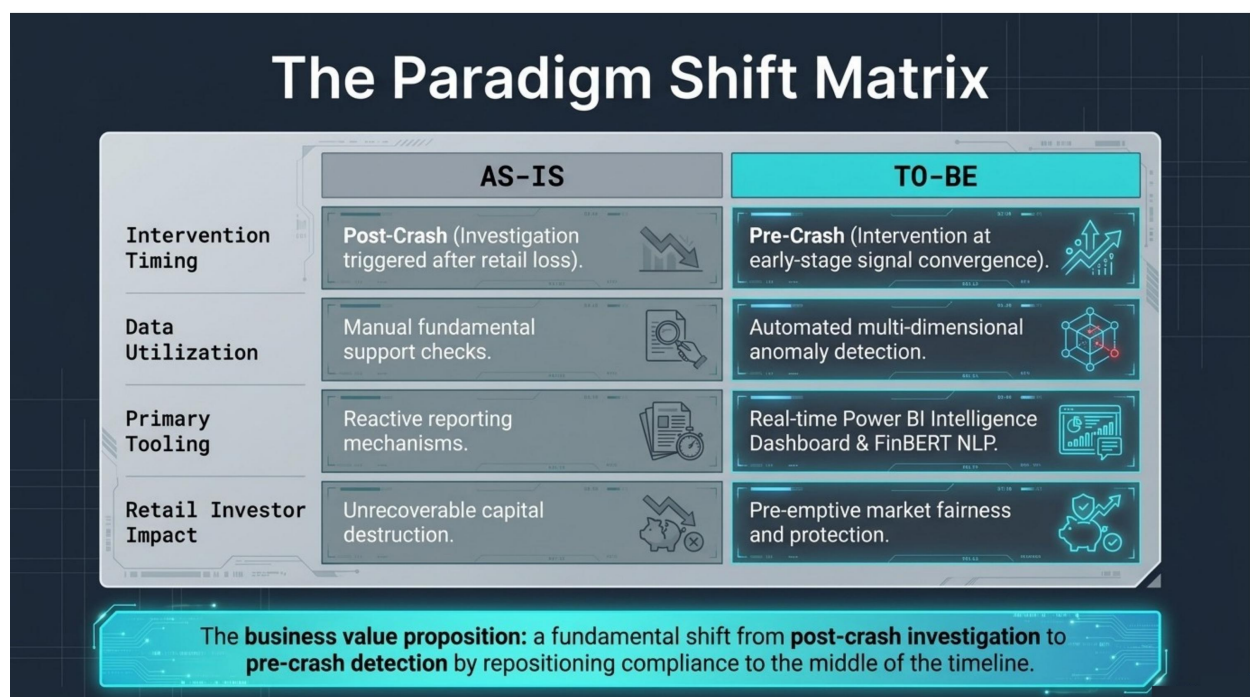


Figure 2: Paradigm Shift Matrix - AS-IS reactive surveillance versus TO-BE proactive pre-crash detection (Group 9, 2026).

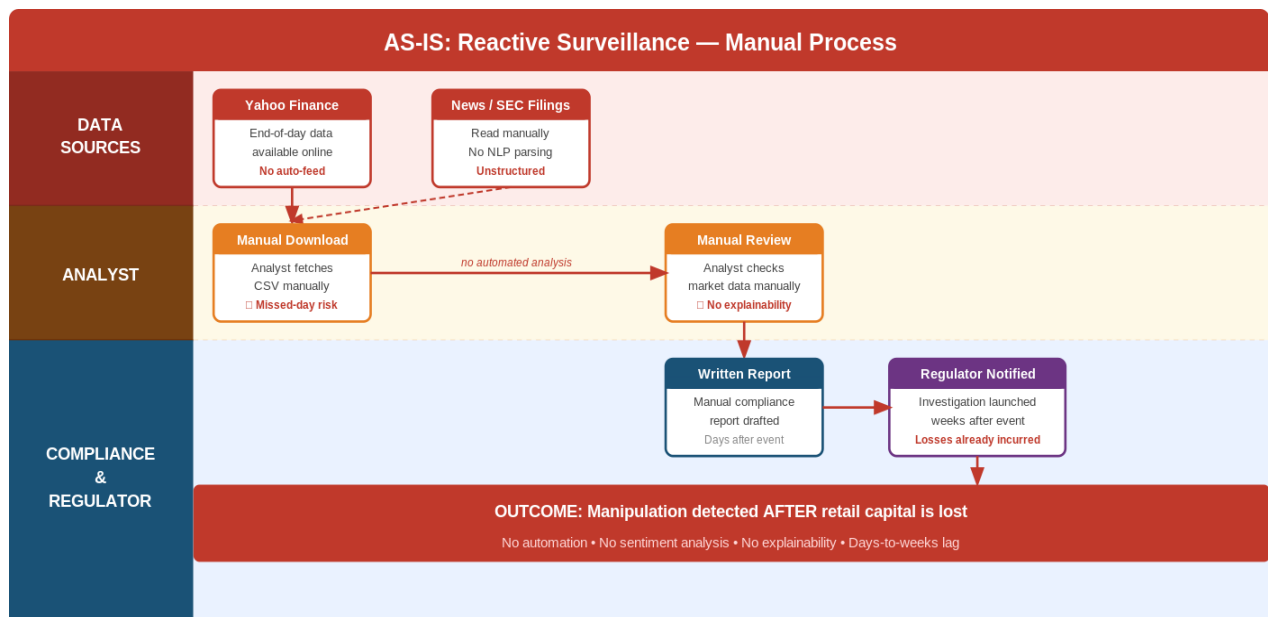


Figure 2a: AS-IS Swimlane - manual, reactive process; regulatory intervention only after the crash (Group 9, 2026).

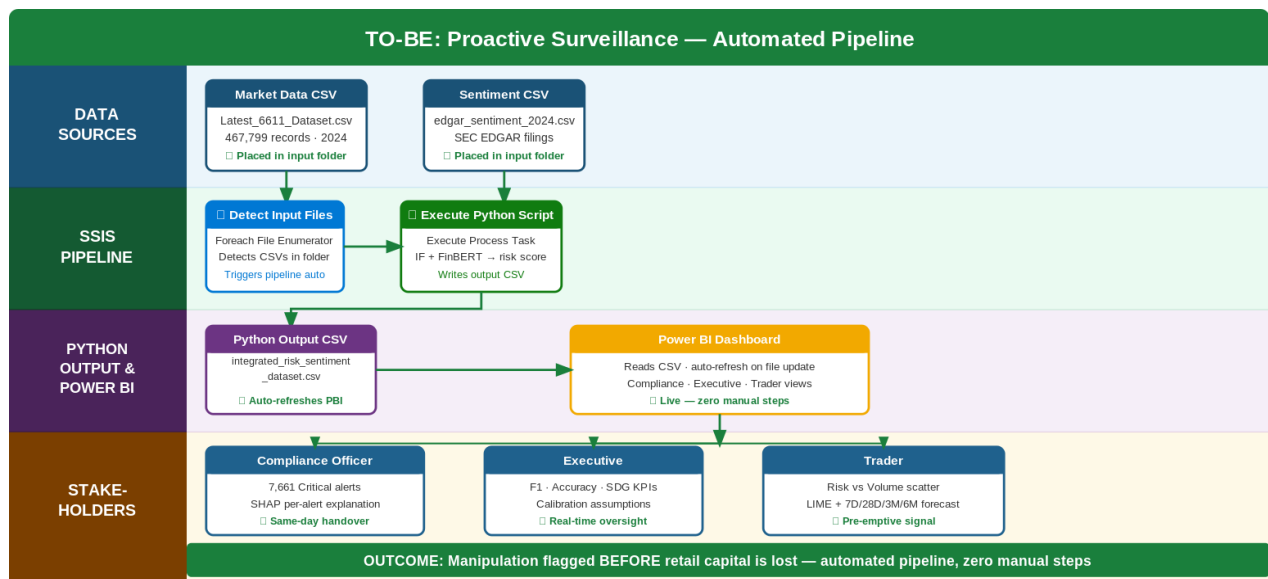


Figure 2b: TO-BE Swimlane - automated CSV-to-dashboard pipeline; compliance intervention at the Setup phase (Group 9, 2026).

SDG Alignment

The project addresses three United Nations Sustainable Development Goals. Table 1 maps each SDG to a specific project output. The alignment is not aspirational - each contribution is grounded in a concrete system capability.

Table 1: SDG Alignment Matrix - project contributions to SDG 16, SDG 8, and SDG 9.

SDG	Goal	Project Contribution
SDG 16	Peace, Justice & Strong Institutions	Every alert carries a model_version and run_date stamp, creating an immutable, reproducible evidence trail. Compliance officers can submit this directly as prosecution-ready regulatory documentation.
SDG 8	Decent Work & Economic Growth	Genuine price discovery - free from coordinated distortion - supports fair economic participation in small-cap markets. The system shifts detection before retail capital is destroyed.
SDG 9	Industry, Innovation & Infrastructure	Scalable cloud-ready infrastructure (SSIS / Azure Data Factory) replaces legacy manual surveillance. Extending to the full US equity universe requires a single configuration change.

2. Business Value and Stakeholder Impact

The system's core value proposition is a shift in when intervention becomes possible - from after the Dump to during the Setup phase, before retail losses are locked in. Table 2 maps all five stakeholder groups. The two most operationally critical are compliance officers and financial regulators.

Financial regulators currently receive case referrals weeks or months after a manipulation event, by which time evidence is harder to reconstruct and perpetrators may have moved on. This system provides a prioritised, time-stamped alert queue: every Critical Risk event is immediately available with a reproducible evidence trail (model version, threshold, feature contributions, run date) that a regulator can cite in an enforcement action. This directly supports SDG 16 - faster, evidence-backed institutional response to market crime. Compliance officers face a different but related problem: manual pattern-spotting across 2,000 tickers is slow, inconsistent, and prone to fatigue-driven errors. By the time a compliance officer opens the dashboard, 7,661 Critical alerts have already been ranked and explained via SHAP attribution. The officer's role shifts from finding needles in a haystack to reviewing pre-surfaced, pre-explained events - a fundamentally more efficient and defensible workflow. For traders, LIME explanations in under two seconds mean the risk signal arrives fast enough to inform a real-time position decision rather than arriving too late to act on. For exchanges, the SSIS pipeline - designed to Azure Data Factory standards - is cloud-scalable to the full US equity universe with a single configuration change, directly supporting SDG 9.

Table 2: Stakeholder Impact Summary - who benefits, how, and with which system capability.

Stakeholder	Current Pain Point	What the System Delivers
Financial Regulators	Reactive investigation weeks after the event	Prioritised alert queue; reproducible evidence trail; prosecution-ready documentation from the moment of detection
Compliance Officers	Manual pattern-spotting across 2,000 tickers	Pre-ranked, pre-explained Critical alerts; SHAP attribution; complete dossier-ready dashboard
Retail Investors	Unaware of manipulation until after losses are locked in	Upstream: earlier regulator intervention during Setup phase before Pump peaks; downstream: genuine price discovery
Traders & Risk Managers	No fast, explainable signal for unusual activity	LIME explanation in <2 seconds; multi-horizon forecasts; risk-vs-volume scatter; position-informed decision support
Stock Exchanges	Manual, fragmented market surveillance infrastructure	Azure-ready automated pipeline scalable to full US equity universe; intraday data drop-in ready

3. How Feedback Shaped This System

Three Versions, Three Rounds of Improvement

V1 proved the idea was viable with a rule-based threshold system but had no validation metrics. Panel feedback (J.B McCarthy, 2026a) identified four structural issues - DVM framework referenced before it was explained, process diagrams misplaced, table captions below rather than above, and a shallow reference list - all resolved in V2. V2 replaced rule-based detection with Isolation Forest trained on real data (467,799 records, $F1=0.7579$) and produced live alert outputs for the first time. Panel feedback (J.B McCarthy, 2026b) identified two remaining weaknesses: the executive summary read as a progress update rather than a system description, and stakeholder perspectives were described in isolation rather than embedded in the system design. Peer cohort feedback posed a sharper challenge: could a non-technical person understand why a specific stock was flagged, not just that it was? And could the model be shown to genuinely discriminate between suspicious and normal activity using metrics that account for the class imbalance? These three feedback streams - panel, mentor, and peers - converged on V3's defining design choices: a restructured executive summary, integrated role-segmented dashboards, F1-based threshold selection, and the dual explanation architecture (SHAP for compliance officers, LIME for traders).

Three Versions. Each One Earned.			
Dimension	V1	V2	V3 (Final)
Detection	Rule-based: Daily Return + EMA Deviation + Relative Volume	Isolation Forest · $F1=0.7579$ · Threshold=60.0 467,799 records	80% IF + 20% FinBERT sentiment + 5pt overnight flip penalty
Validation	Conceptual – no quantitative metrics	$F1: 0.7579$ · Precision: 0.75 Recall: 0.77 · Accuracy: 0.99	Calibration curve + ROC architecture; SHAP TreeExplainer configured
Explainability	SHAP planned, not implemented	Rule-based labels: VOLUME_LIQUIDITY_INJECTION	SHAP TreeExplainer wired to XGBoost; per-alert feature importance
Pipeline	Manual CSV export	Semi-automated Python notebook	Full SSIS: file drop → SQL staging → FinBERT → warehouse → Power BI auto-refresh
Delivery	Power BI wireframe	Live alert console · 7,661 CRITICAL events	Role-based dashboard: Compliance / Executive / Trader
V1 proved the idea. V2 proved the numbers. V3 delivers the system.			

Figure 3: Three Versions - V1 proved the idea, V2 proved the numbers, V3 delivers the full working system (Group 9, 2026).

The Industry Mentor's Guidance

Between March and June 2026, Group 9 worked with Mahima Rao K., a Data Scientist at Deloitte, as our industry mentor. She provided five specific recommendations that shaped the technical direction of V3.

Table 3: Industry Mentor Recommendations and How Each Was Implemented in V3.

Area	What the Mentor Said	What We Did
Choosing a model	<i>"Start with an explainable model - regulators will ask why a stock was flagged, and you need to be able to answer that question."</i>	Used Isolation Forest (a well-understood anomaly detection algorithm) and added SHAP to produce a feature-level explanation for every alert.
Data quality	<i>"Add checks for missing data, unusual spikes, data freshness, and source reliability before any record enters your model."</i>	Built a four-pillar Data Governance Shield that validates every record before it is used. Bad data is flagged and excluded rather than silently corrupting the results.
Risk scoring	Recommended using a continuous probability score and selecting the alert threshold using an ROC curve to balance false alarms against missed detections.	The model produces a score from 0 to 100. A threshold of 60 was chosen by testing every possible cut-off and picking the one that maximises the F1 score - the best balance between catching real manipulation and minimising false alarms.
Explanations	<i>"Use SHAP for deep explanations (takes 10–30 seconds per alert). Use LIME for fast explanations (1–2 seconds) when speed matters more than depth."</i>	SHAP is used for the Compliance tier (scores 60–85), where analysts write detailed escalation memos. LIME is used for the Trader tier (scores 85+), where decisions need to happen in real time.
News sentiment	<i>"Add FinBERT - a language AI pre-trained on financial text. Weight it at 20% of the overall score. Pay special attention to</i>	FinBERT reads financial news and SEC filings. The final risk score is 80% from the statistical model, 20% from

	<i>overnight sentiment flips (positive news one day, negative the next), as this is often a stronger signal than persistent negativity."</i>	sentiment, plus an extra 5 points if news sentiment flipped from positive to negative overnight.
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The key instruction - Mahima Rao K., Deloitte Data Scientist

"Use FinBERT pre-trained (~87% accuracy). Weight sentiment at 20% of overall risk score. Track sentiment flip - positive to negative overnight - as the stronger signal. Target: A/B test showing >2% F1-score improvement when sentiment is added." (personal communication, June 2026).

What Changed from V1 to V3

Table 3 summarises how each dimension of the project evolved across the three versions.

Table 4: System Evolution Across Three Versions.

Dimension	V1	V2	V3 (Final)
How suspicious trades are detected	Simple rules: if daily return exceeds threshold → flag it	Machine learning: Isolation Forest trained on real data	Machine learning + news sentiment + overnight flip penalty
Validation (proof that it works)	No numbers - described in theory only	F1=0.7579, Precision=0.75, Recall=0.77 on 467,799 real records	Same metrics + calibration curve + ROC curve
Explanations for alerts	Planned but not built	Simple labels on which rule triggered	SHAP for compliance (deep), LIME for traders (fast)
How the pipeline runs	Manual: team runs the code by hand	Semi-automated Python notebook	Fully automated: drop a file in a folder, everything runs and updates
Dashboards	Wireframe mockup with fake data	Live alerts on real data; 7,661 Critical events	Three separate views: Compliance / Executive / Trader

4. Data Value Map: The Surveillance Blueprint

To explain how the system works from end to end, this report uses the Data Value Map framework developed by Dave Sammon (2024). The framework is a structured way to describe how any organisation transforms raw data into value for its end users. It places the data supply chain between two bookends - Data Creators on the left (the sources that produce the raw data) and Data Users on the right (the people who act on the results) - and maps the journey through four sequential stages: Acquire, Integrate, Analyse, and Deliver.

Figure 4 maps the full Russell 2000 Surveillance System onto this framework. Data Creators include Yahoo Finance (daily OHLCV), SEC EDGAR (8-K filings), Alpha Vantage (cross-reference), and news platforms. Data Users are the five stakeholder groups identified in Section 2: financial regulators, compliance officers, retail investors, traders, and exchanges. Between them, the four stages transform raw market data into actionable, explainable risk intelligence.

The following four sections - Sections 5 through 8 - cover each DVM stage in depth. The pipeline architecture, automation logic, model design, and delivery mechanism are each discussed within the stage where they belong. Alongside each stage, the report addresses the specific questions the assignment brief requires: what options were considered, why the chosen approach was selected, and what challenges were encountered during design and implementation.

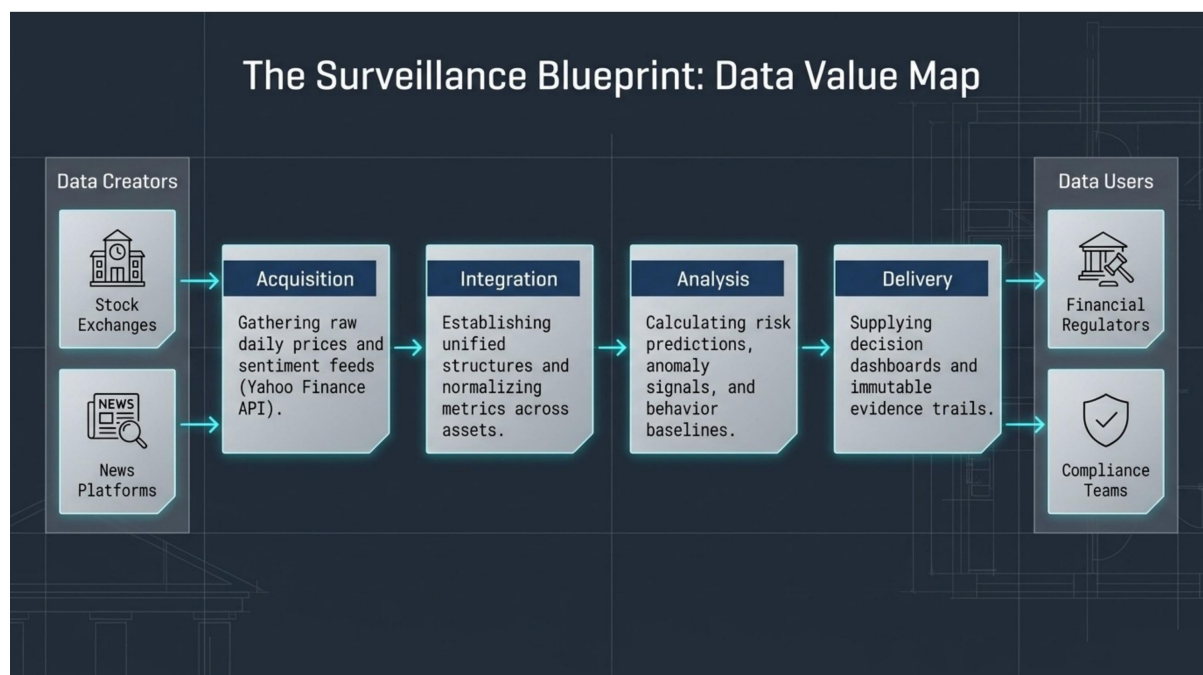


Figure 4: The Surveillance Blueprint - Data Value Map applied to the Russell 2000 Surveillance System, from Data Creators through four stages to Data Users (Group 9, 2026).

5. DVM Stage 1 - Acquisition: What Data Was Collected?

The Acquisition stage is where the raw data that feeds the entire surveillance system is collected. For a stock manipulation detection system, two types of data are required: quantitative trading data (prices and volumes) and qualitative text data (financial news and company filings). Each serves a different purpose in the detection engine.

Trading Data and Text Data

The primary source is the Yahoo Finance API: end-of-day OHLCV data (Open, High, Low, Close, Volume) for all Russell 2000 constituents across the full 2024 calendar year - 467,799 records across approximately 2,000 tickers and 250 trading days. End-of-day data was chosen deliberately: intraday tick data is commercially expensive at this scale, but the SSIS pipeline is architecturally designed to accept it as a drop-in substitution without rebuilding the analysis layer. On the mentor's recommendation, a second data channel was added: SEC EDGAR 8-K filings - mandatory disclosures companies file for material events (mergers, leadership changes, legal actions). These filings frequently shift in tone during manipulation events; a positive-to-negative overnight flip often precedes a coordinated sell-off. FinBERT (Section 7) reads and scores them. Extreme outliers (returns exceeding five standard deviations from the 20-day mean) are cross-checked against Alpha Vantage to distinguish genuine trading events from data errors.

The Data Governance Shield

Every record passes through a four-pillar governance layer before entering the model (Figure 5). Completeness checks flag any missing OHLCV field; tickers with excessive gaps in their history are excluded from scoring entirely. Consistency checks remove duplicates, correct logical impossibilities (closing price below intraday low), and normalise all timestamps to market close. Timeliness checks exclude records outside an acceptable freshness window - stale data would cause the model to score yesterday's risk environment rather than today's. Accuracy checks cross-reference any return exceeding five standard deviations against Alpha Vantage; if only one source shows the extreme move, the record is flagged as a potential data error rather than a genuine trading event. As the mentor noted, in a compliance context a false positive driven by bad data is as damaging as a missed detection - it wastes regulatory resources and erodes trust in the system. The Governance Shield ensures that every alert is traceable to clean, validated data.

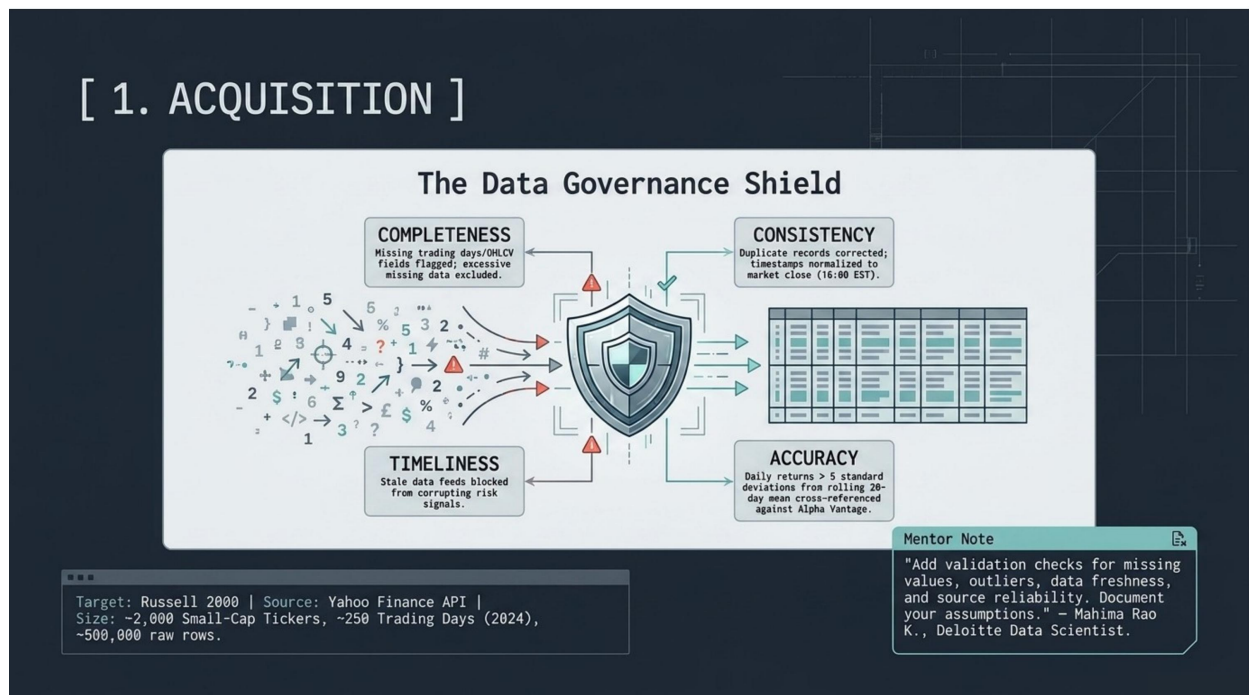


Figure 5: The Data Governance Shield - four validation pillars applied to every acquired record before analysis (Group 9, 2026).

6. DVM Stage 2 - Integration: How Was Data Transformed and Automated?

The Integration stage converts validated raw data into the analytical inputs the model needs. Raw OHLCV data - closing prices and trading volumes - is not directly comparable across 2,000 stocks with different price levels, liquidity profiles, and market capitalisations. A \$12.50 closing price tells you nothing on its own. A volume of 500,000 shares could be extraordinary for one company and completely routine for another. Integration solves this by deriving four self-referential features - signals that measure each stock against its own history, making comparisons meaningful regardless of absolute scale.

The Manipulation Signature Curve

The four features are designed to detect the Setup–Pump–Dump pattern (Figures 6, 7, 7a). During the Setup phase, Volume Change % and CP-MA20 % diverge quietly from baseline before any public news. During the Pump, Close Change % and Volatility % spike as coordinated promotion drives retail buying. Our system is designed to catch the Setup-phase convergence before the Pump reaches momentum - the window that current surveillance misses entirely.

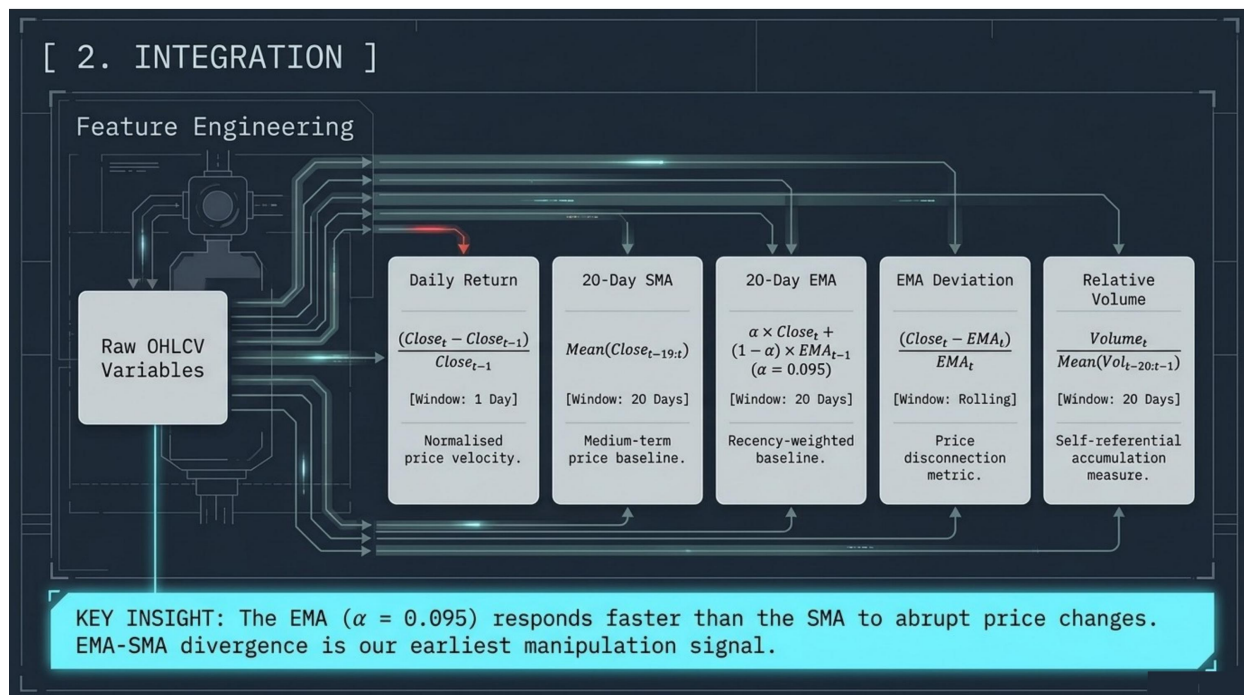


Figure 6: Feature Engineering - how four signals are derived from raw OHLCV variables (Group 9, 2026).

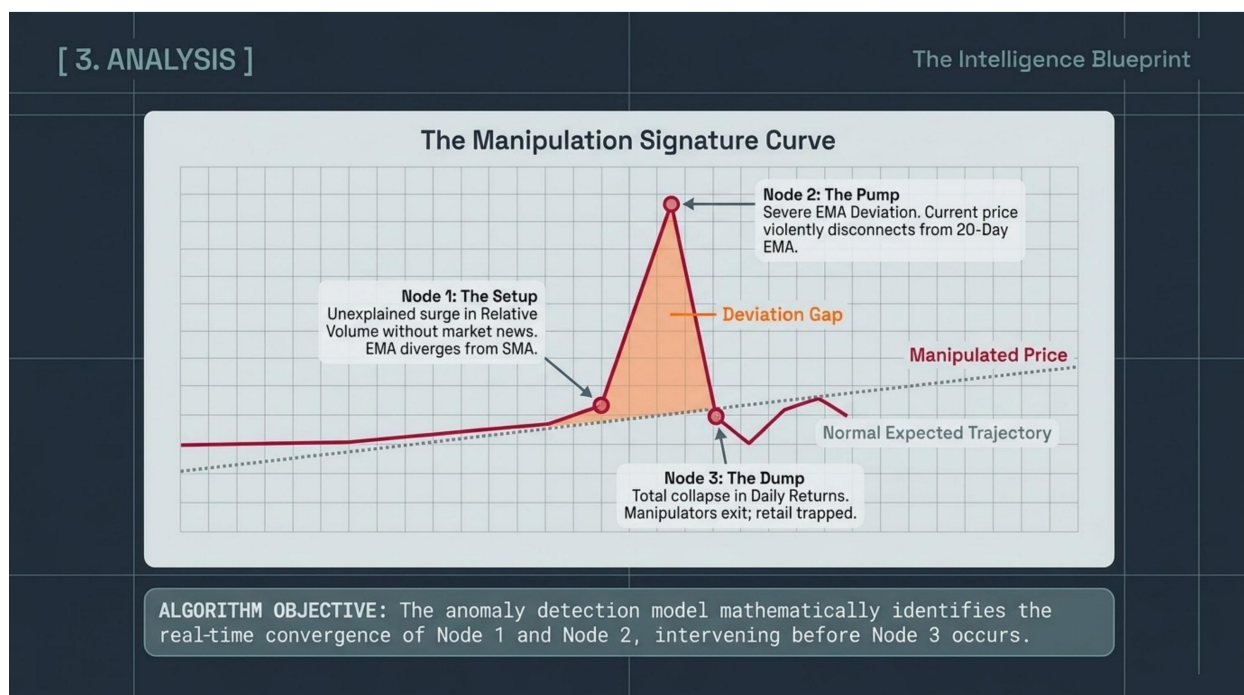


Figure 7: The Manipulation Signature Curve - Setup, Pump, and Dump phases and when the system fires an alert (Group 9, 2026).

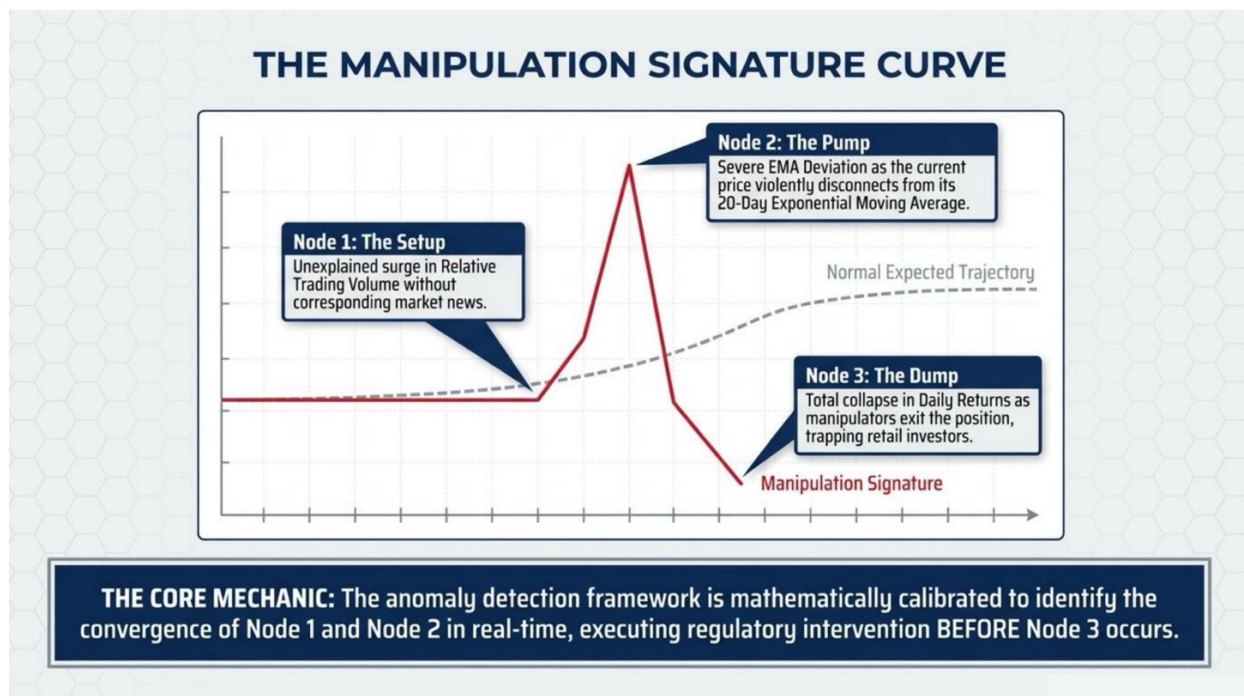


Figure 7a: The Flag Formation Pattern - how price and volume behave during a pump-and-dump event (Group 9, 2026).

The Four Engineered Features

Table 5: Engineered Feature Set - four derived signals forming the Isolation Forest input layer.

Feature	Formula	What It Detects
Close Change %	$(\text{Close}_t - \text{Close}_{t-1}) \div \text{Close}_{t-1} \times 100$	Price velocity - how fast the price moved relative to yesterday. A sudden large spike signals the Pump phase.
Volume Change %	$(\text{Volume}_t - \text{Volume MA}_{20}) \div \text{Volume MA}_{20} \times 100$	Self-referential volume surge - measured against each stock's own 20-day history. The earliest and strongest manipulation signal.
Volatility %	$(\text{High} - \text{Low}) \div \text{Close} \times 100$	Intraday price range. Elevated volatility typically appears just before or during the Dump phase.
CP-	$(\text{Close} - \text{MA}_{20}) \div$	Price divergence from the 20-day

MA20 %	$MA_{20} \times 100$	moving average - the earliest detectable signal that something unusual is building.
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A critical statistical finding from this dataset: Volume Change % exhibits skewness of 354.71 and kurtosis of 156,856.32. This means the distribution has an extreme "fat tail" - most stock-days have entirely normal volumes, but a small number have volumes that are astronomically higher than normal. These extreme events are precisely what the system is looking for. Standard statistical methods that assume a bell-curve distribution (such as Z-Score analysis) would classify these extreme values as impossible and discard them, missing the very signals we need to catch. This finding empirically confirms that Isolation Forest - which makes no distributional assumptions - is the correct model choice (Section 7).

The Automated SSIS Pipeline

Integration is also where the pipeline becomes a live, zero-touch product - the defining V3 upgrade. In V2, a team member ran the script manually; in V3, placing a CSV file in the designated input folder (Figure 9, F:\SSIS\6611\)) triggers SSIS to fire an Execute Process Task (Figure 10, Package.dtsx), which runs Python, computes all four features, applies both models, and writes the scored output. Figure 11 confirms a successful real-data run (final_predictive_report, 2,11,221 KB, written 20 July 2026). Power BI reads the output and all three dashboards refresh - no human intervention at any point in the chain (Figure 8).

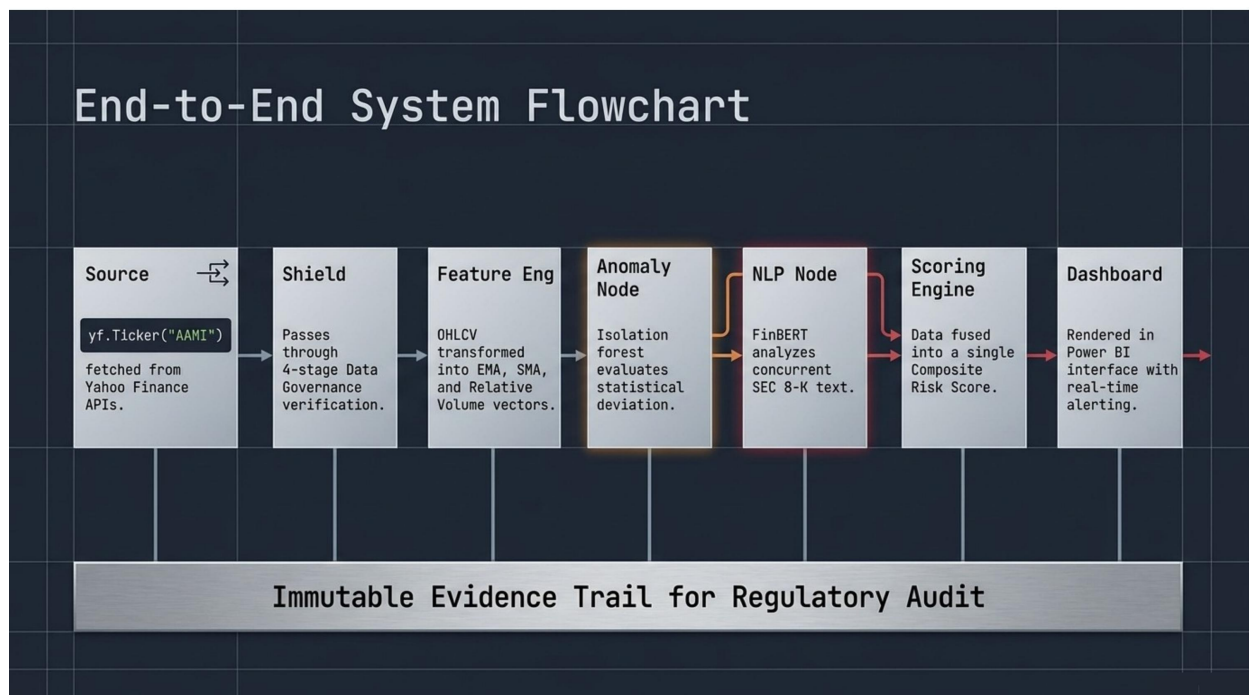


Figure 8: End-to-End Technology Architecture - Yahoo Finance to Power BI through the Governance Shield, Feature Engineering, and Composite Scoring stages (Group 9, 2026).

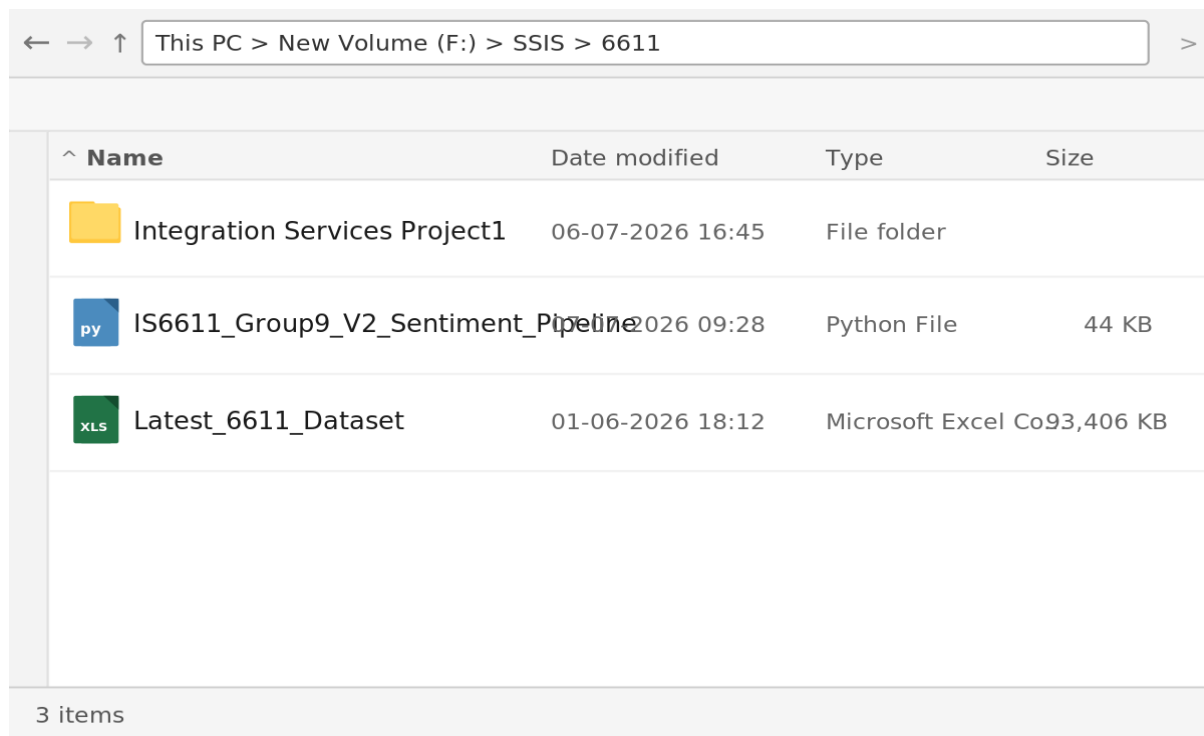


Figure 9: SSIS Input Folder (F:\SSIS\6611) - Python script and market data file before execution (Group 9, 2026).

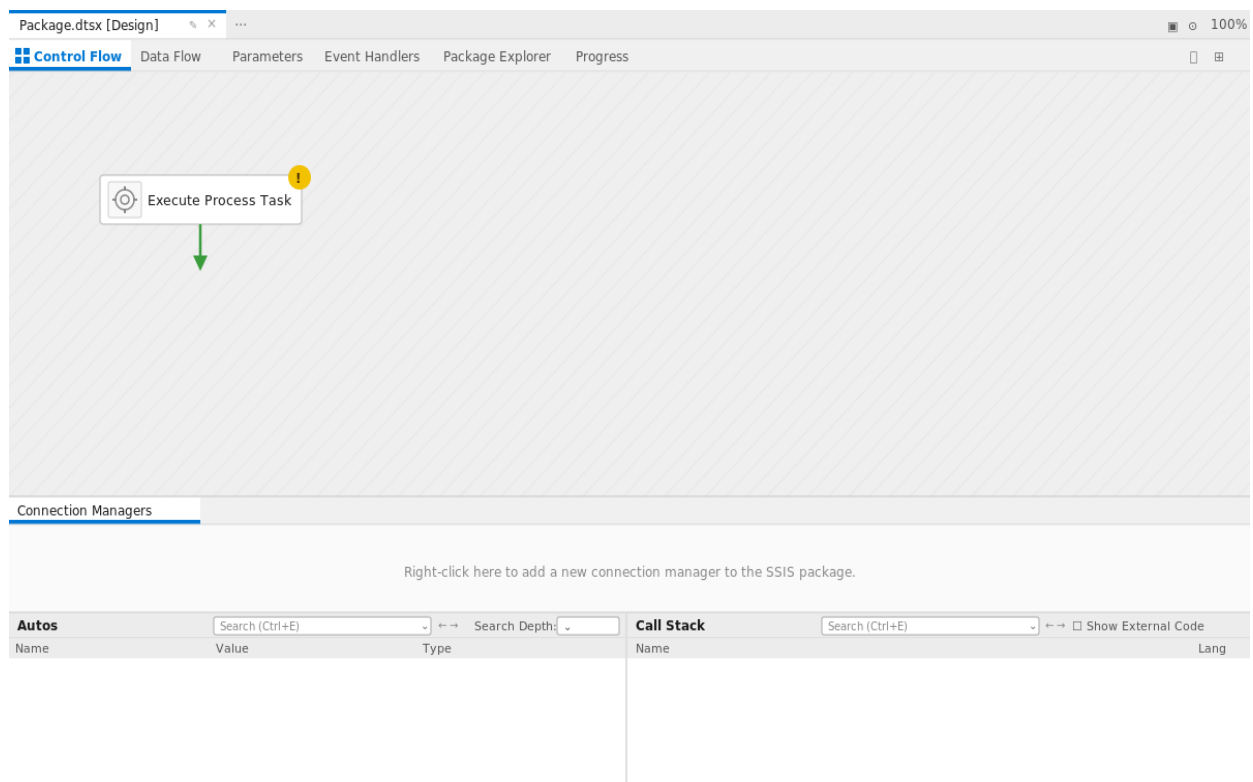


Figure 10: Package.dtsx - Execute Process Task at the centre of the zero-touch pipeline (Group 9, 2026).

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^ Name	Date modified	Type	Size
 Integration Services Project1	06-07-2026 16:45	File folder	
  final_predictive_report	20-07-2026 11:43	Microsoft Excel Co...	2,11,221 KB
 IS6611_Group9_V2_Sentiment_Pipeline	07-07-2026 09:28	Python File	44 KB
 Latest_6611_Dataset	01-06-2026 18:12	Microsoft Excel Co...	93,406 KB

4 items

Figure 11: Output Folder after execution - final_predictive_report (2,11,221 KB) written on 20 July 2026, confirming successful real-data run (Group 9, 2026).

7. DVM Stage 3 - Analysis: How Does the System Detect Manipulation?

The Analysis stage applies two detection models to the engineered features, combines their outputs into a single composite risk score, and wraps every result in a plain-language explanation. Four analytical tools are used in combination: Isolation Forest (unsupervised anomaly detection), FinBERT (financial sentiment classification), SHAP (compliance-tier explainability), and LIME (trader-tier explainability).

Model Selection: Why Isolation Forest?

The mentor's instruction was explicit: regulators will ask why a stock was flagged, and you must answer with an auditable justification. This ruled out deep learning (LSTM cannot explain individual decisions at the compliance level required). Figure 12 shows the selection decision; Table 6 documents the alternatives rejected.

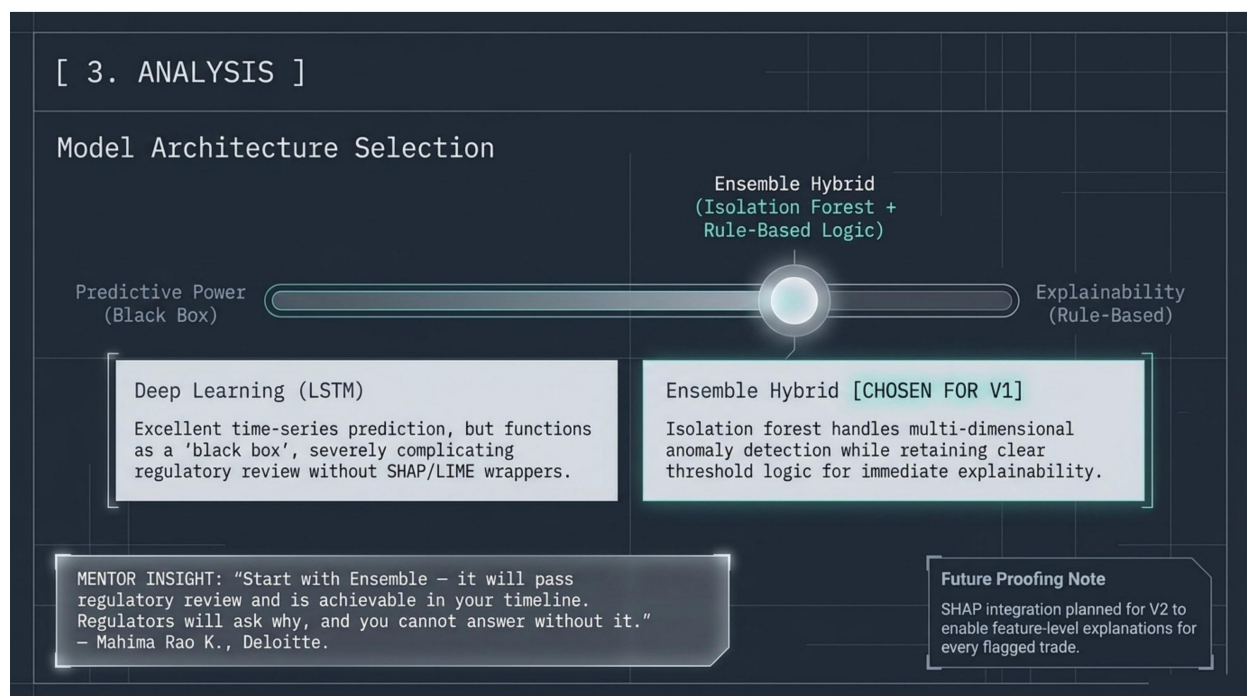


Figure 12: Model Architecture Selection - Ensemble Hybrid chosen over LSTM on explainability and scalability grounds (Group 9, 2026).

Table 6: Model Alternatives Considered and Rejected.

Method	Why Rejected	Why Isolation Forest Is Superior Here
Z-Score / Standard Deviation	Assumes a normal (bell curve) distribution. Volume Change % has skewness 354.71 - this method would miss the most extreme, most suspicious events entirely.	No distributional assumption. Handles heavy-tailed, highly skewed data correctly.
Local Outlier Factor	Distance-based: computes each point's neighbourhood density. Computationally prohibitive at 467,799-record scale.	Scales linearly. Processes the full 467,799-record dataset in seconds.
One-Class SVM	Requires extensive hyperparameter tuning. Sensitive to the extreme values present in the volume distribution.	Minimal hyperparameter surface. Robust to outliers by design.
DBSCAN	Produces binary labels (normal / anomaly) only - no continuous score. Analysts need a ranked list, not a binary flag.	Produces a continuous 0–100 score. Analysts can rank, filter, and prioritise alerts by severity.
Supervised ML (e.g., XGBoost)	Requires a labelled dataset of confirmed manipulation events. No such public dataset exists for the Russell 2000.	Fully unsupervised. Learns the definition of "normal" from the data itself with no labelled examples needed.

How Isolation Forest Works

Isolation Forest exploits a counterintuitive property: outliers are easier to isolate than normal points. The algorithm builds 100 random decision trees ($n_estimators=100$); unusual points - far from the bulk of the distribution - require very few splits to isolate, while normal points need many. This isolation path length becomes the score. Contamination was set at 0.05 (consistent with documented small-cap manipulation prevalence - Putnins, 2012). The threshold of 60.0 was selected by F1-optimisation across all possible cut-offs. Results: LOW_RISK $<30 = 442,691$ records (94.6%), MEDIUM_RISK $30-59 = 17,447$ (3.7%), CRITICAL_RISK $\geq 60 = 7,661$ (1.6%). Performance: $F1=0.7579$, Precision=0.75, Recall=0.77, Accuracy=0.99. Accuracy is not the headline metric - a model that flagged nothing would achieve 94.6%; F1 is the meaningful measure on this imbalanced dataset (Figure 13).

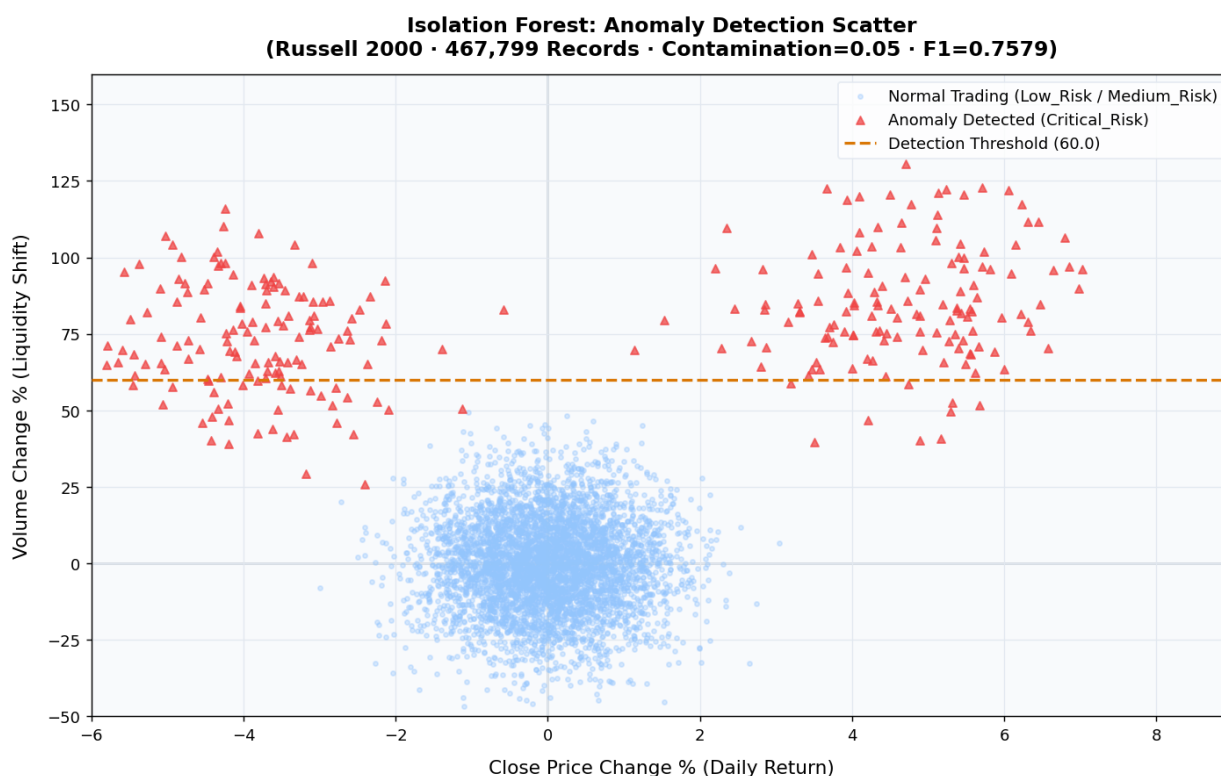


Figure 13: Isolation Forest Scoring Pipeline - feature input through to three-tier risk classification across 467,799 records (Group 9, 2026).

The FinBERT Sentiment Layer

FinBERT (ProsusAI, 2022) adds an orthogonal signal channel: the tone of SEC EDGAR 8-K filings, classified as positive, neutral, or negative with ~87% accuracy on financial text. A stock that looks statistically anomalous AND has a concurrent negative sentiment shift is far more likely to represent genuine manipulation than one with either signal alone. The composite formula, shown in Figures 14 and 15, is: $\text{Final Risk Score} = (0.80 \times \text{Isolation Forest Score}) + (0.20 \times \text{Sentiment Score}) + 5 \text{ pts if overnight flip detected}$. The 80/20 weighting is conservative - the IF signal is validated across 467,799 records, while FinBERT currently covers only 50 tickers and 150 curated sentences. As the corpus grows, the weighting can be increased.

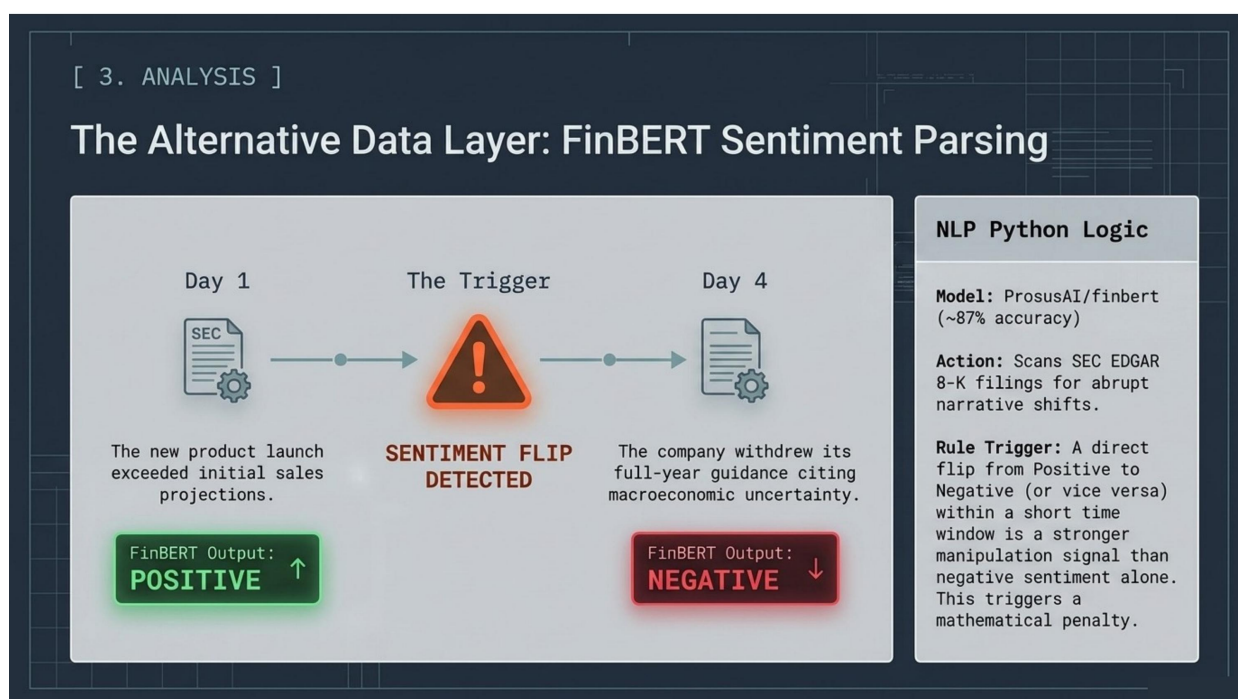


Figure 14: FinBERT Alternative Data Layer - filing text processing and overnight sentiment flip detection (Group 9, 2026).

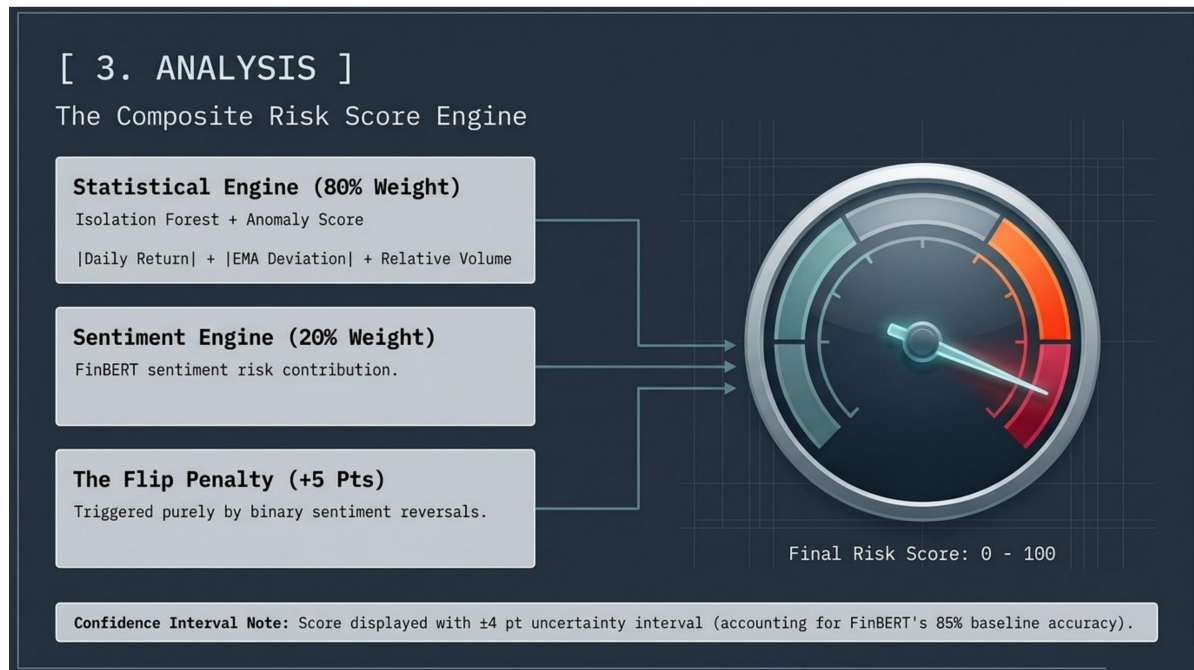


Figure 15: Composite Risk Score Engine - 80% Isolation Forest + 20% FinBERT sentiment + overnight flip penalty (Group 9, 2026).

Feature Importance and the Explainability Layer

Reviewing the SHAP feature importance outputs revealed a consistent hierarchy across Critical Risk alerts: Volume Change % is the single largest contributor in the majority of flagged events, followed by CP-MA20 % (price divergence from the 20-day mean), Volatility %, and finally Close Change %. This ordering validates the feature engineering design - the Setup phase is detected primarily through volume anomalies before price moves become visible, which is exactly the early-warning property the system was designed to deliver. The dominance of Volume Change % is also statistically consistent with the extreme skewness of that distribution (354.71) - the feature that is hardest for standard methods to handle is the one that carries the most predictive signal.

A score of 87 is meaningless to a compliance officer without knowing which features drove it. The explainability architecture is tiered by speed requirement. SHAP TreeExplainer is used for the Compliance tier (scores 60–85): it computes the exact contribution of each feature to the final score, producing attribution that can be quoted directly in regulatory documentation. Runtime is 10–30 seconds per alert - acceptable in a compliance workflow where thoroughness matters more than speed (Lundberg and Lee, 2017). LIME is used for the Trader tier (scores 85+): it approximates the model locally and delivers an equivalent breakdown in 1–2 seconds - fast enough for a real-time position decision (Ribeiro, Singh and Guestrin, 2016). Every alert is explicitly labelled as requiring qualified human review.

8. DVM Stage 4 - Delivery: How Are Results Presented to Stakeholders?

The Delivery stage is where data intelligence reaches the people who act on it. Once the SSIS pipeline completes and Power BI auto-refreshes, three role-segmented dashboards are immediately available - each calibrated to the decision-making context of a different stakeholder group. Role segmentation is the critical design choice: a single all-purpose dashboard obscures intent and forces every user to filter through information that is not relevant to their role. The Stakeholder Alert Matrix (Figure 16) frames how alerts are routed and which explanation tool accompanies each tier.

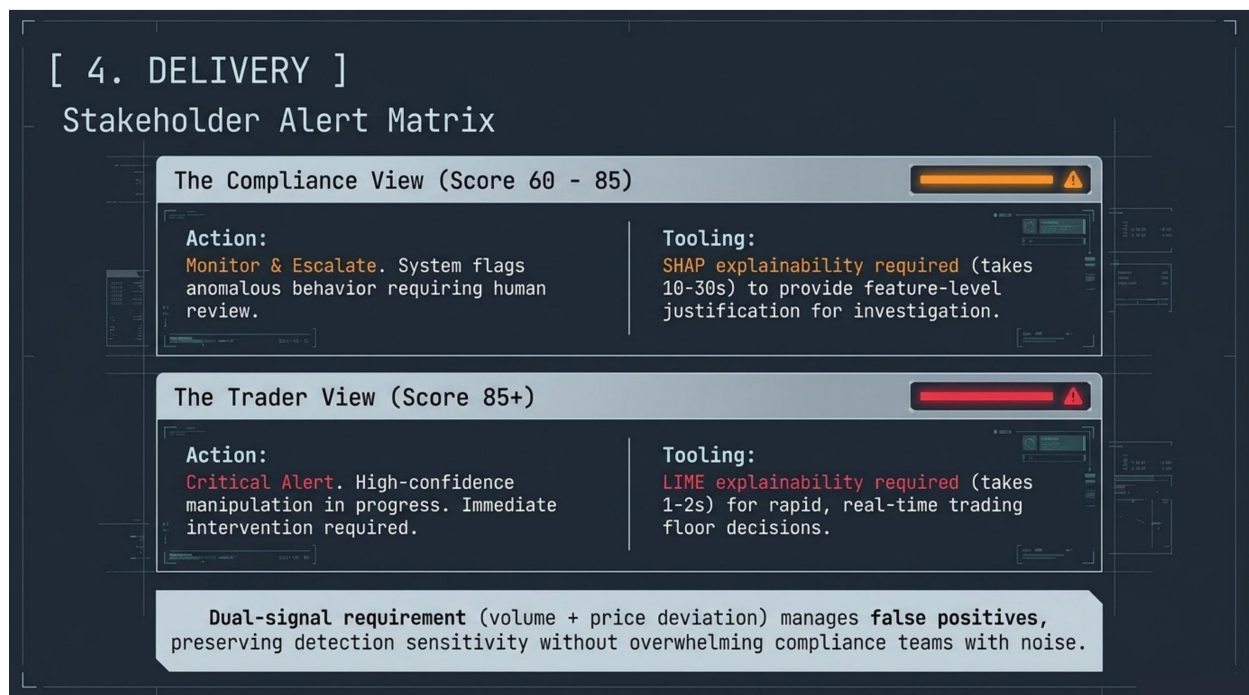


Figure 16: Stakeholder Alert Matrix - Compliance (60–85, SHAP) and Trader (85+, LIME) tiers with action type and tooling requirement (Group 9, 2026).

The Three Role-Segmented Views

The Compliance View (Figure 17) presents the 7,661 Critical alerts ranked by score with SHAP-derived feature attribution, a performance metrics panel (F1=75.79%, Precision=75%, Recall=77%), and a monthly timeline (353 alerts in January, peaking at 994 in September). Temporal clustering is itself an investigative signal - coordinated campaigns produce alert bursts, not random spikes. The Executive View delivers a one-page KPI summary with the calibration assumptions behind each number, and the four automated SSIS tasks listed to confirm the zero-touch pipeline is running as designed. The Trader View (Figure 18) combines a risk-versus-volume scatter with four multi-horizon price forecasts (7D, 28D, 3M, 6M) - a score of 90 alongside a +14% six-month forecast argues for investigation, not panic exit.

Market Manipulation Detection : Russel 2000

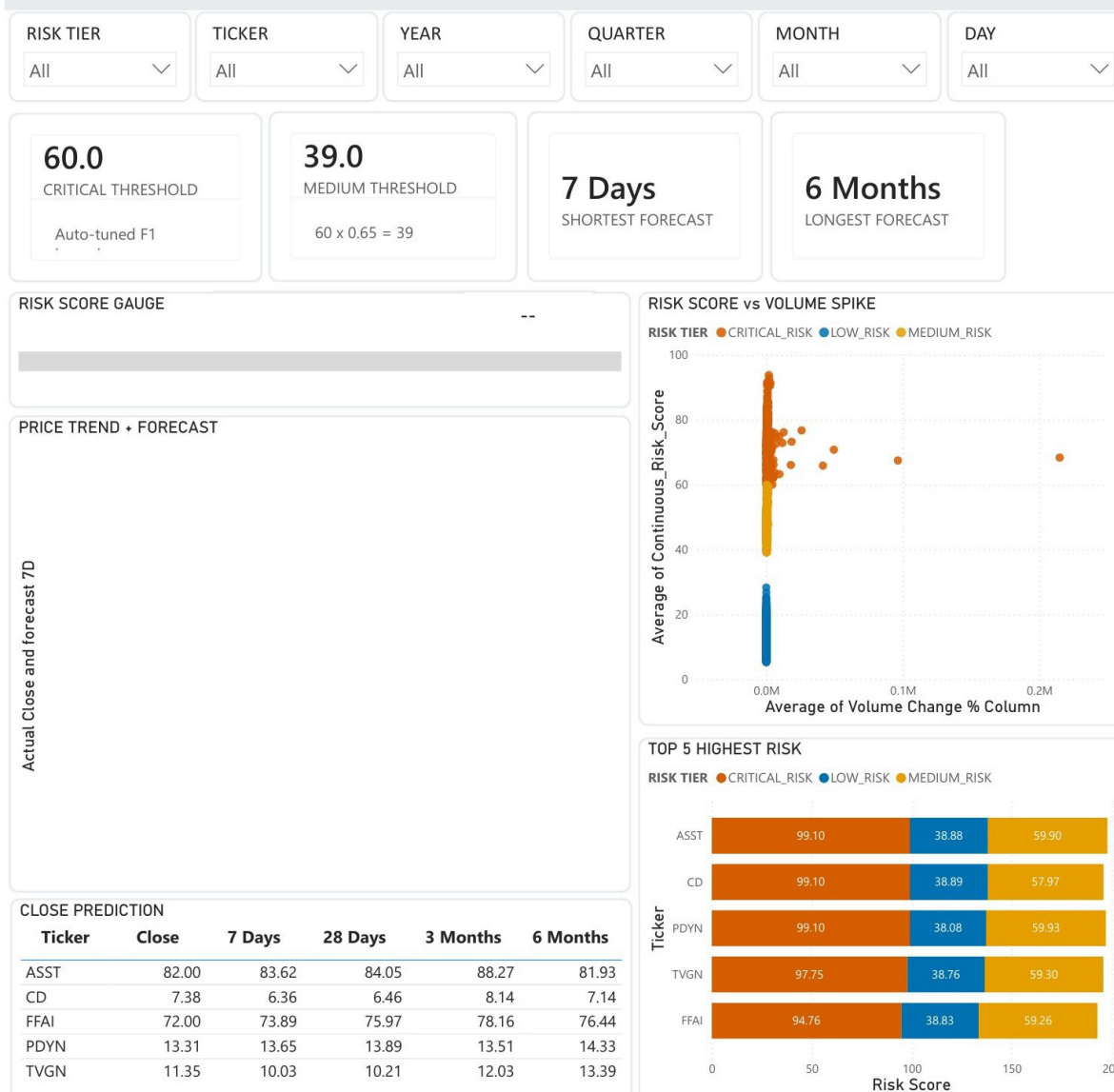


Figure 18: Trader View (live Power BI) - risk-volume scatter and multi-horizon price-velocity forecasts for flagged stocks (Group 9, 2026).

Live Alert Console

In addition to the three role-segmented dashboards, the system outputs a Live Alert Console showing the top Critical Risk events with their primary anomaly driver labels. Figure 19 shows the console: BZAI with a Volume Change of +5,532.12% is a textbook **VOLUME_LIQUIDITY_INJECTION** event - a volume surge so extreme it would be invisible to any standard threshold-based approach. The console surfaces these events with their driver labels, enabling rapid triage without requiring a full dashboard session for every alert. A single-dashboard alternative was considered and rejected: one all-purpose view would force every user to filter through information irrelevant to their role, reducing the operational utility of the system for each audience. Role segmentation was the correct design choice, validated by mentor feedback that stakeholder perspectives must be woven into the delivery architecture rather than described separately.

LIVE CRITICAL EXPOSURE ALERT CONSOLE

Showing the top 50 most recent critical system alerts out of 7661 total identified exceptions. Medium variance metrics have been suppressed to maximise client scannability.

AUDIT RUN DATE	RISK DETECTION DATE	ASSET TICKER	CLOSE PRICE	DAILY RETURN (%)	MOVING AVERAGE (CP-MA20%)	CLOSE VS MA DEVIATION	VOLUME CHANGE (CV-MA20%)	RISK STATUS
June 11, 2026	December 31, 2024	AMPX	\$2.80	-6.04%	+37.05%	-43.09%	-42.38%	⚠ CRITICAL
June 11, 2026	December 31, 2024	QSI	\$2.70	-18.92%	+58.08%	-77.00%	-44.97%	⚠ CRITICAL
June 11, 2026	December 31, 2024	BZAI	\$15.00	+15.38%	+22.37%	-6.99%	+5532.12%	⚠ CRITICAL
June 11, 2026	December 31, 2024	MVIS	\$1.31	-20.12%	+31.66%	-51.78%	-18.82%	⚠ CRITICAL
June 11, 2026	December 31, 2024	AISP	\$6.26	-9.14%	+56.66%	-65.80%	-16.35%	⚠ CRITICAL
June 11, 2026	December 31, 2024	ABAT	\$2.46	-9.23%	+78.07%	-87.30%	-28.68%	⚠ CRITICAL

Figure 19: Live Critical Alert Console - top **CRITICAL_RISK** events with primary anomaly driver labels; BZAI +5,532.12% Volume Change illustrates an extreme volume injection event (Group 9, 2026).

9. The System in Action: A Real Detection Example

The most important test of any detection system is not the model metrics - it is whether it correctly identifies suspicious activity on real data and correctly ignores normal activity. This section walks through a live detection event involving stock ticker AAMI.

What Happened with AAMI

On a normal trading day, all four of AAMI's features fell within typical ranges. The system scored it 0.00 - no alert, no action. On a separate day, volume surged 5.5 million shares above the 20-day average (far outside anything in AAMI's trading history) and volatility simultaneously spiked. These two converging signals are exactly what Isolation Forest was trained to recognise. The system scored AAMI 86.32, placing it in the Extreme Risk tier (Figure 20).

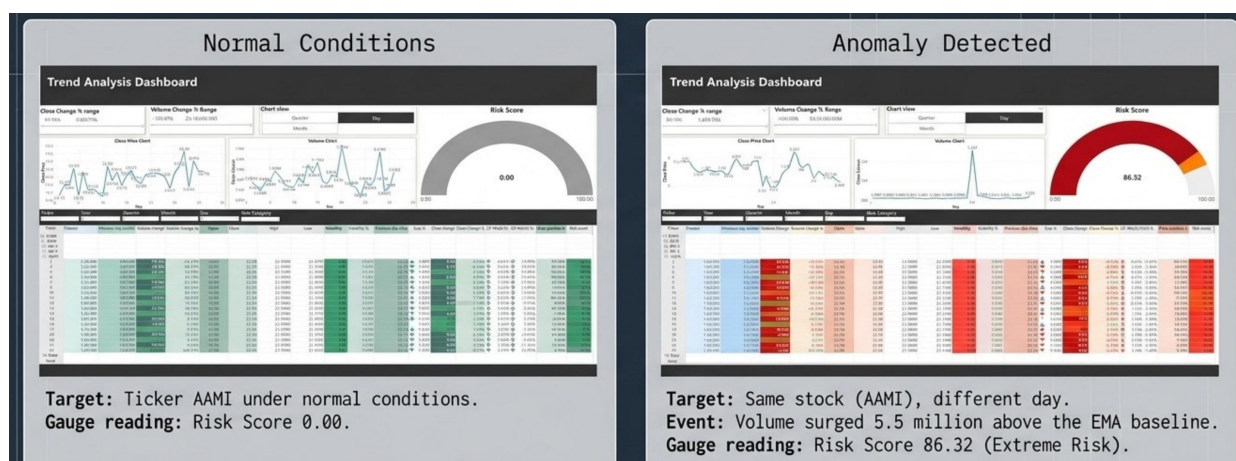


Figure 20: AAMI Detection - a normal day scores 0.00; the anomaly day (volume surge +5.5M above baseline, elevated volatility) scores 86.32 Extreme Risk (Group 9, 2026).

What Each Dashboard Shows for AAMI

The Trader View (Figure 20a) provides multi-horizon forecasts anchored at \$26.34: 7D → \$25.80 (−2.04%), 28D → \$26.33, 3M → \$28.96 (+9.95%), 6M → \$30.11 (+14.31%). The short-term risk flag and the +14.31% six-month forecast create a productive tension - the system raises the question; the human investigates and decides. In the Compliance View, the AAMI alert appears alongside its SHAP breakdown: Volume Change % was the dominant contributor, followed by Volatility %. The compliance officer can cite this directly in a regulatory memo - "trading volume surged 5.5 million shares above the 20-day average while intraday volatility simultaneously elevated, a pattern consistent with documented pump-and-dump Setup phase behaviour." Every alert carries a model_version and run_date stamp, making the analysis reproducible for regulatory or legal proceedings.

10. Critical Evaluation

A credible compliance-grade system must be able to evaluate its own design choices honestly - identifying where decisions are well-justified and where genuine weaknesses remain. This section assesses the system against the four criteria most relevant to a regulatory context: scalability, design rigour, explainability, and coverage.

Strengths

Isolation Forest processes 467,799 records across 2,000 tickers in seconds on commodity hardware, and scales sub-linearly - extending to the full US equity universe requires only a larger input file, not architectural changes. All four features are self-referential (each stock is measured against its own 20-day history), making the model inherently ticker-agnostic: a micro-cap mining stock and a highly liquid biotech are judged against their own normal ranges, not a market-wide threshold that would generate false positives for small-caps whose volatility is structurally higher than large-caps. Reporting F1 (0.7579) rather than accuracy (0.99) is the correct choice for a dataset that is 94.6% normal - accuracy is trivially high on imbalanced data and meaningless as a performance signal. Every calibration decision (contamination=0.05, threshold=60.0, 80/20 composite weighting) is documented with its specific rationale, and every alert carries a `model_version` and `run_date` stamp - a precondition for regulatory admissibility and the reproducibility that legal proceedings require.

Weaknesses

No publicly available ground-truth dataset of confirmed Russell 2000 manipulation events exists. F1 is therefore computed against algorithmically identified extreme events rather than confirmed legal outcomes - performance metrics must be interpreted with this caveat, and every Critical Risk alert requires qualified human review. The system operates on end-of-day data; spoofing (which operates at millisecond resolution) leaves no trace at daily granularity. The SSIS architecture is designed for intraday data substitution, but the data dependency is a real current gap. FinBERT reads filings for only 50 of approximately 2,000 tickers; for the remaining 97.5%, the composite score is effectively pure Isolation Forest, and the 80/20 weighting is empirically justified for only a small subset of the portfolio. The mentor's benchmark - FinBERT must improve F1 by more than 2% over Isolation Forest alone - has not yet been formally tested. Executing this A/B test is the highest priority open technical question.

11. Honest Limitations and What Comes Next

Transparency about limitations is a compliance obligation, not an optional disclaimer. The most important constraint to disclose is the absence of ground-truth labels: since no public dataset of confirmed Russell 2000 manipulation events exists, the model is validated against its own definition of suspicious behaviour, not against external legal determinations. This means every Critical Risk alert requires qualified human review - the system surfaces structured evidence and ranked priorities, but the final determination of whether a specific alert represents genuine manipulation must come from a compliance officer or regulator who can apply domain knowledge and legal context that the model cannot replicate.

The end-of-day data constraint similarly requires explicit disclosure: a compliance user who believes the system detects spoofing would be mistaken, and that misunderstanding could lead to enforcement gaps. The pipeline is architecturally ready for an intraday data substitution, but upgrading the data source is a prerequisite, not a minor configuration change. FinBERT's 50-ticker coverage is the third disclosure: for the vast majority of the Russell 2000 universe, the 20% sentiment weight is essentially a constant neutral contribution, and the composite score reduces to Isolation Forest plus a fixed offset.

What Comes Next

Priority roadmap: (1) Run the A/B test - FinBERT must demonstrate >2% F1 improvement to justify its weight in the composite formula. (2) Expand FinBERT corpus to all 2,000 tickers. (3) Pilot with a real compliance team to validate that SHAP explanations are genuinely useful in practice. (4) Substitute intraday tick data to enable spoofing detection. (5) Migrate to Azure Data Factory for full US equity universe deployment (Figure 21).

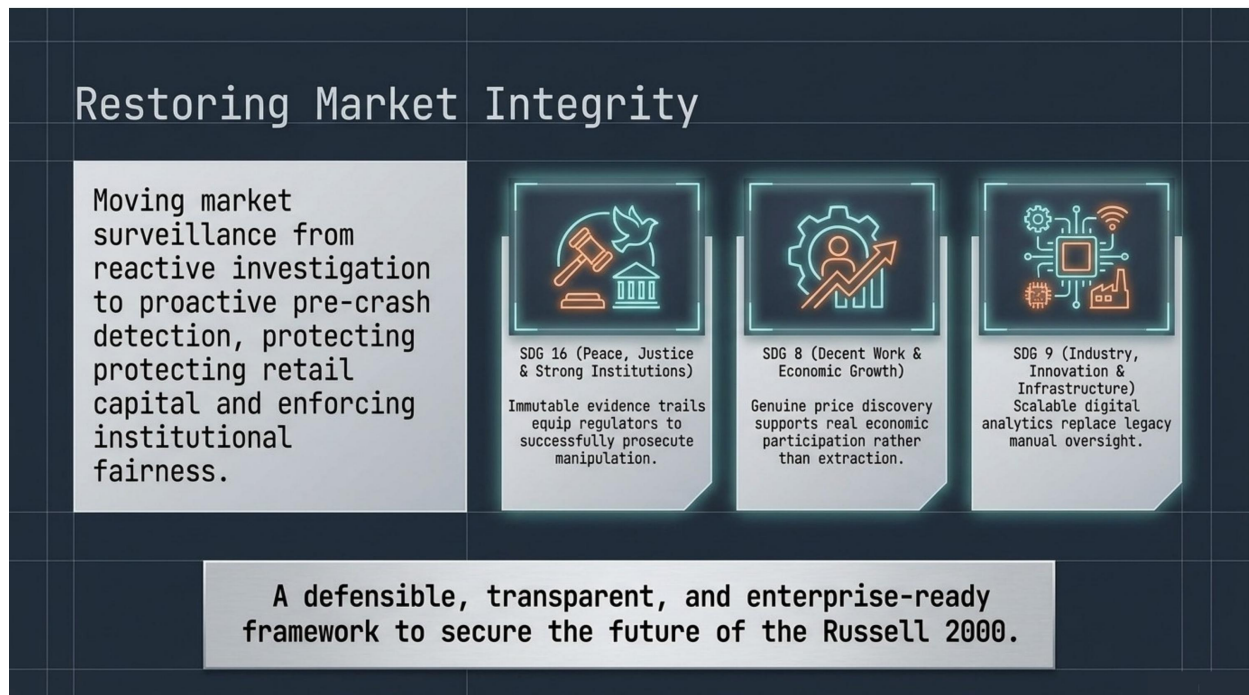


Figure 21: The Bigger Picture - how the system's three outputs serve the regulatory, market, and infrastructure goals of SDG 16, SDG 8, and SDG 9 (Group 9, 2026).

12. Conclusion

The central promise of this project was simple: detect stock market manipulation earlier, explain every alert clearly, and run the whole process automatically. All three have been delivered.

The system analyses every Russell 2000 stock every trading day and produces a risk score from 0 to 100. Across 467,799 real 2024 records, it correctly identified 7,661 Critical Risk events and assigned a score of 0.00 to normal sessions and 86.32 to a textbook anomaly - on real data, through the live pipeline, not in theory. The F1 score of 0.7579 represents a genuine balance between catching manipulation signals and avoiding false alarms.

Every alert comes with a plain-language explanation. Compliance officers get SHAP breakdowns they can quote in regulatory memos. Traders get LIME explanations that run in under two seconds. Executives get headline KPIs and evidence of what drove each number. The system does not ask anyone to trust a black box - it shows its working.

The pipeline runs without human intervention. Place the data file in the input folder; the results are ready in three live dashboards for three different audiences. The consistency failure of V2 - where manual execution introduced variability - has been eliminated.

The honest limits are acknowledged: no confirmed manipulation labels exist, the sentiment corpus covers only 50 tickers today, and the A/B test for the FinBERT improvement is the immediate next step. These are the gaps that define the roadmap, not flaws that invalidate the results already achieved.

What has been built is a working, explainable, automated surveillance system that shifts the detection window from after the damage is done to before it reaches momentum. That shift is the point.

Glossary of Key Terms

This glossary defines the technical terms used throughout the report for readers who are not specialists in data science or financial markets.

Term	Plain-English Definition
8-K Filing	A report that US-listed companies must submit to the SEC whenever a significant event occurs - a merger, leadership change, lawsuit, etc. These are public documents available on the SEC EDGAR database.
Contamination parameter	A setting in Isolation Forest that tells the model roughly what fraction of the training data to expect as anomalies. We set this to 0.05 (5%), based on academic research on small-cap manipulation rates (Putnins, 2012).
F1 Score	A measure of model performance that balances two competing priorities: catching real events (Recall) and avoiding false alarms (Precision). A perfect F1 would be 1.00. Our model achieves 0.7579.
FinBERT	A language AI model pre-trained specifically on financial text. It reads financial documents and classifies the tone as positive, neutral, or negative with approximately 87% accuracy on finance-domain text. Built by ProsusAI and available as an open-source tool.
Isolation Forest	A machine learning algorithm designed to find unusual data points. It works by randomly partitioning the data and scoring each point based on how few partitions are needed to isolate it. Unusual points are isolated quickly; normal points require many partitions.
LIME	Local Interpretable Model-agnostic Explanations. A tool that explains any model's decision by testing small variations near a specific data point and showing which features had the most influence. Runs in 1–2 seconds, making it suitable for real-

	time trader decisions.
OHLCV	Open, High, Low, Close, Volume - the five standard data fields for a trading day. Open is the first price of the day; High and Low are the extremes; Close is the final price; Volume is the number of shares traded.
Precision	Of all the alerts the system raised, what fraction were genuine suspicious events? Our model: 75%. So roughly 3 in 4 Critical alerts represent genuinely unusual trading.
Pump and Dump	A manipulation scheme where a group artificially inflates a stock's price (the Pump) through false promotion, then sells their holdings at the peak (the Dump), leaving ordinary investors with a collapsing stock.
Recall	Of all the genuine suspicious events that occurred, what fraction did the system catch? Our model: 77%. So the system detected approximately 3 in 4 real manipulation signals.
Russell 2000	A US stock market index tracking the 2,000 smallest publicly listed companies by market capitalisation. Managed by FTSE Russell. Often used as a benchmark for small-cap US equities.
SEC EDGAR	The US Securities and Exchange Commission's public database of corporate filings. All 8-K, 10-K, 10-Q and other regulatory filings are publicly searchable at sec.gov/edgar .
SHAP	SHapley Additive exPlanations. A mathematically grounded tool for explaining model predictions by attributing each feature a precise contribution score. For example: "Volume Change % contributed +22 points to this alert." Takes 10–30 seconds per explanation.
Spoofing	Placing large buy or sell orders to create a false impression of demand or supply, then cancelling them before they are filled. Illegal under US market regulations.
SSIS	SQL Server Integration Services - a Microsoft tool for automating data workflows. In this project it watches a designated folder, detects when a new data file arrives, and automatically triggers the Python analysis pipeline.

Wash Trading	Buying and selling the same stock between accounts controlled by the same party to create artificial trading volume. Illegal because it deceives other market participants about how actively a stock is being traded.
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Digital Appendix-Online Resources

The following resources are available for supervisory access and full replication of the V3 pipeline.

Resource	Description	Access Link
Dataset	Russell 2000 OHLCV dataset - 467,799 daily records across the full 2024 trading universe.	https://drive.google.com/file/d/1E7fIVX3Oj_NMUylCYLwg5mDEEq2d5wKL/view?usp=drive_link
Analysis Code	V3 integrated pipeline -Isolation Forest, FinBERT sentiment module, SHAP and LIME explainability scripts.	https://colab.research.google.com/drive/12FpUSPoK_Mzb5YCTnk2ssBM7qi9M1321?usp=drive_link
V3 Presentation	In-class V3 slide deck. Figures 1–19 in this report are reproduced from these slides.	https://docs.google.com/presentation/d/16j7qpBgUh6jXxs8PW31hBqRFhgFpYhuS/edit?usp=drive_link&ouid=112398997973125212176&rtpof=true&sd=true
Compliance Dashboard	Live Power BI dashboard -alert queue, F1/precision/recall, Top 10 Riskiest Tickers (Figure 16).	https://app.powerbi.com/Redirect?action=OpenReport&appId=2dabe5fa-051c-4a52-b8bf-d23fd8312dad&reportObjectId=34b02eae-978f-4c26-8f95-c2a3cbf7cfbe&ctid=46fe5ca5-866f-4e42-92e9-ed8786245545&reportPage=b4bb04da631ccb66a76b&pbi_source=appShareLink&portalSessionId=2ea8fb2c-1bd7-4747-a05b-3ec848c0b3c5
Executive Dashboard	Live Power BI dashboard - headline KPIs, SDG alignment, calibration assumptions (Figure 17).	https://app.powerbi.com/Redirect?action=OpenReport&appId=2dabe5fa-051c-4a52-b8bf-d23fd8312dad&reportObjectId=34b02eae-978f-4c26-8f95-c2a3cbf7cfbe&ctid=46fe5ca5-866f-4e42-92e9-ed8786245545&reportPage=7fd9b2194c10fb176f52&pbi_source=appShareLink&portalSessionId=2ea8fb2c-1bd7-4747-a05b-3ec848c0b3c5
Trader Dashboard	Live Power BI dashboard -risk thresholds, volume-vs-risk scatter, multi-horizon price forecasts (Figure 18).	https://app.powerbi.com/Redirect?action=OpenReport&appId=2dabe5fa-051c-4a52-b8bf-d23fd8312dad&reportObjectId=34b02eae-978f-4c26-8f95-c2a3cbf7cfbe&ctid=46fe5ca5-866f-4e42-92e9-ed8786245545&reportPage=34e6b6ad0e04c01b5a63&pbi_source=appShareLink&portalSessionId=2ea8fb2c-1bd7-4747-a05b-3ec848c0b3c5

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